

Issues With Dario's Collinear Counterterms for the Two-Loop $q\bar{q} \rightarrow HHH$ Amplitude

pygloop qqbar_nX implementation notes

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Author-Correction Note

After this note was first written, Dario Kermanschah confirmed that the denominator displayed in the published Eqs. (12)–(13) contains a typo: the auxiliary propagators must be built from the adjacent gluon momenta k_{g_1} and k_{g_2} , not from the light-bridge momentum k_1 . In the notation used below, the corrected denominators are therefore

$$A_\xi(k_{g_i}) = (k_{g_i} - \xi)^2 - \xi^2,$$

not $A_\xi(k_1)$. The current `qqbar_nX` implementation has been updated accordingly by splitting the relevant collinear gluon edge and routing the new massive auxiliary segment as $k_{g_i} - \xi$. The severe soft-collinear enhancement diagnosed in this note is consequently best read as a diagnosis of the pre-correction $k_1 - \xi$ implementation, not of the corrected counterterm.

Chapter 1

Context and Setup

The process considered here is the two-loop amplitude

$$d(p_1) \bar{d}(p_2) \longrightarrow H(p_3) H(p_4) H(p_5),$$

where the three Higgs bosons couple to a massive top-quark loop and the massive loop is connected to the external massless quark line by two gluons. The local initial-state collinear counterterms are those of Kermanschah’s construction in Ref. [1]. The implementation discussed here is the one in the `qqbar_nX` process of `pygloop`.

The DOT file used for the representative topology plots is the final-state symmetrized process generated by `qqbar_nX`. It contains four original graphs, grouped into two canonical topologies, and the two ISR collinear counterterms attached to each original graph. The figures were produced from a GammaLoop state in which this DOT file was loaded and then exported with `save dot`; running `just draw` in that exported state produced the PDFs included in this note.

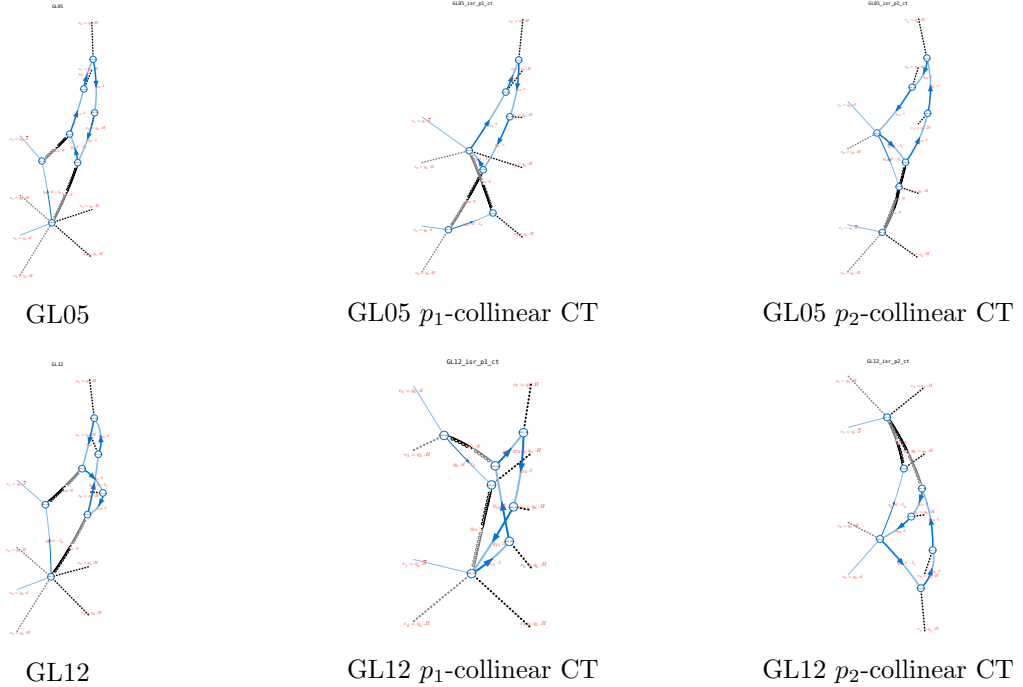


Figure 1.1: Two representative final-state-symmetrized original topologies and their local ISR collinear counterterms. The full GammaLoop drawing grid is included in Appendix A.

Chapter 2

Four-Dimensional Local Representation

2.1 Light-line denominators

Only the light massless bridge is needed to expose the endpoint problem. We write the loop momentum on the light quark bridge as

$$k_1 \equiv K_0,$$

with the two adjacent gluon momenta denoted by k_{g_1} and k_{g_2} . For the scaling discussion we use the convention

$$k_{g_1} = k_1 - p_1, \quad k_{g_2} = -k_1 - p_2.$$

The sign convention is not important for the power counting; what matters is that k_{g_1} becomes longitudinal and on shell in the p_1 -collinear soft endpoint, while k_{g_2} becomes longitudinal and on shell in the p_2 -collinear soft endpoint.

Suppressing all massive-loop denominators, the relevant original light part is

$$D_{\text{orig}} = k_1^2 k_{g_1}^2 k_{g_2}^2.$$

The exact- ξ counterterm topology replaces the non-local light denominator by the auxiliary denominator

$$A_\xi(k_1) \equiv (k_1 - \xi)^2 - \xi^2 = k_1^2 - 2\xi \cdot k_1.$$

The two counterterm light denominator structures used in the DOT files are

$$D_{C_1} = k_1^2 k_{g_1}^2 A_\xi(k_1), \quad D_{C_2} = k_1^2 A_\xi(k_1) k_{g_2}^2.$$

The helper external momenta are only used to realize A_ξ as an ordinary GammaLoop denominator. In the exact- ξ DOT strategy, the original graphs have the helper half-edges marked as dummy. In the C_1 counterterm only the $Q(2), Q(4)$ helper pair is active, and in the C_2 counterterm only the $Q(3), Q(5)$ helper pair is active. The inactive pair is dummy. The mass override of the shifted auxiliary edge is ξ^2 , represented by the corresponding active helper momentum squared, so the construction remains ξ -generic.

2.2 Numerators and the observed mismatch

The original numerator contains the massless bridge spinor chain

$$N_{\text{orig}} = \bar{v}(p_2) \gamma_{\mu_2} \not{k}_1 \gamma_{\mu_1} u(p_1) \mathcal{H}^{\mu_1 \mu_2}(k_{g_1}, k_{g_2}; \{p_H\}),$$

where $\mathcal{H}^{\mu_1\mu_2}$ is the massive top-loop tensor with three Higgs insertions. The counterterm numerators are inserted as graph-global tensor network numerators in the DOT file; edge and vertex numerators inside the CT topology are set to one. No finite routing-ratio factors are inserted in the numerator. The relevant collinear-projected part of the first CT is schematically

$$N_{C_1} \sim \bar{v}(p_2) \Gamma_{C_1}^{\mu_2}(p_1, p_2, k_1, \xi) u(p_1) k_{g_1}^{\mu_1} \mathcal{H}_{\mu_1\mu_2}(k_{g_1}, k_{g_2}; \{p_H\}),$$

with an analogous expression for C_2 containing $k_{g_2}^{\mu_2}$. The crucial point is that the original spinor chain has an explicit soft k_1 , while the projected CT numerator can remain finite in the soft-collinear endpoint unless a Ward identity kills $k_{g_1}^{\mu_1} \mathcal{H}_{\mu_1\mu_2}$ or $k_{g_2}^{\mu_2} \mathcal{H}_{\mu_1\mu_2}$.

2.3 p_1 -soft-collinear scaling

Take

$$p_1 = E(1, 0, 0, 1), \quad p_2 = E(1, 0, 0, -1), \quad \xi = p_1 + p_2 = (2E, 0, 0, 0),$$

and approach the p_1 -soft-collinear endpoint with

$$k_1 = E\lambda(1, 1, 0, 1), \quad \lambda \rightarrow 0^+.$$

Then

$$k_1^2 = -E^2\lambda^2, \tag{2.1}$$

$$k_{g_1}^2 = (k_1 - p_1)^2 = -E^2\lambda^2, \tag{2.2}$$

$$k_{g_2}^2 = (-k_1 - p_2)^2 = E^2(4\lambda - \lambda^2) = \mathcal{O}(\lambda), \tag{2.3}$$

$$A_\xi(k_1) = k_1^2 - 2\xi \cdot k_1 = -4E^2\lambda + \mathcal{O}(\lambda^2). \tag{2.4}$$

Hence

$$D_{\text{orig}} \sim \lambda^5, \quad D_{C_1} \sim \lambda^5, \quad D_{C_2} \sim \lambda^5.$$

The denominators alone do not distinguish the original from the singular CT. The mismatch is instead

$$N_{\text{orig}} \sim \lambda, \quad N_{C_1} \sim \lambda^0 \quad \text{if} \quad k_{g_1}^{\mu_1} \mathcal{H}_{\mu_1\mu_2} \not\rightarrow 0.$$

Therefore the direct four-dimensional local objects scale as

$$|\mathcal{I}_{\text{orig}}| \sim \lambda^{-4}, \quad |\mathcal{I}_{C_1}| \sim \lambda^{-5}, \quad |\mathcal{I}_{C_2}| \sim \lambda^{-4},$$

with the total dominated by C_1 . The p_2 -soft-collinear endpoint is the beam-exchanged version:

$$|\mathcal{I}_{\text{orig}}| \sim \lambda^{-4}, \quad |\mathcal{I}_{C_1}| \sim \lambda^{-4}, \quad |\mathcal{I}_{C_2}| \sim \lambda^{-5}.$$

The direct `pygloop` four-dimensional evaluator, using the same graph-level numerators and denominator definitions as the emitted DOT file, finds the following fitted powers:

Approach	original	C_1	C_2	total
p_1 -soft-collinear	-4.000	-5.000	-4.000	-5.000
p_2 -soft-collinear	-4.000	-4.000	-4.999	-4.999

where the entries mean $|I| \propto \lambda^p$. This confirms that the problem is already present before any CFF subtleties are introduced.

Chapter 3

Three-Dimensional CFF Representation

3.1 Schematic CFF weight

After applying the causal-flow/energy-residue representation, a local three-dimensional contribution has the schematic form

$$W(\vec{K}_0, \vec{K}_1) = J(\vec{K}_0, \vec{K}_1) \sum_{\omega \in \Omega} \frac{N_{\omega}(\vec{K}_0, \vec{K}_1)}{\prod_{F \in \mathcal{F}_{\omega}} F_{\omega}(\vec{K}_0, \vec{K}_1)}.$$

Here J is the mapping Jacobian and the F_{ω} are the causal energy-surface denominators for a given orientation ω . The exact power of an endpoint scan depends on which cube variable is used to approach the surface and on the selected loop-momentum basis. What is robust in the observed scans is the hierarchy: the CT assigned to the soft-collinear beam is one power worse than the corresponding original graph.

For the endpoint scan used in the GammaLoop-rich-output and custom-CFF checks, the cube variables were taken as

$$x(\Delta) = (\Delta, 0.8941729073613068, 1 - \Delta, 0.05523624797827975, 0.7717783989563521, 0.4505175649210359), \quad \Delta \rightarrow 0^+. \quad (3.1)$$

With threshold subtraction disabled, the observed three-dimensional GammaLoop/CFF weights scale as

$$|W_{\text{orig}}| \sim \Delta^{-1}, \quad |W_{C_i}| \sim \Delta^{-2}, \quad |W_{\text{tot}}| \sim \Delta^{-2},$$

for the CT C_i associated with the beam being approached. The powers are softer than the direct four-dimensional density because the CFF residues and the unit-cube mapping already include part of the spatial measure and energy Jacobian. They nevertheless reproduce the same structural failure: the local CT has one more inverse endpoint power than the original contribution it is supposed to subtract.

3.2 Denominator origin of the residual growth

Near the soft-collinear endpoint, the CFF energy surfaces generated by the light bridge inherit the same small scales as the four-dimensional denominators:

$$E_{k_1} \sim \Delta, \quad E_{k_{g_1}} \sim \Delta, \quad A_{\xi} \sim \Delta.$$

The original graph contains the soft bridge numerator suppression from k_1 . The CT topology has the same small CFF surfaces after the auxiliary replacement, but its projected numerator

contains the longitudinal massive-loop contraction $k_{g_i} \cdot \mathcal{H}$. If that contraction is finite instead of soft, the CFF weight keeps one additional inverse power. This is the three-dimensional version of Eqs. (2.3) and (2.3).

Chapter 4

Ward-Identity Rescue Attempt

4.1 Why the Ward identity should help

The reason a careful routing could in principle cure the problem is the massive-loop Ward identity. For a gluon with momentum k_g attached to a fermion line with momentum q ,

$$k_g = (\not{q} + \not{k}_g - m_t) - (\not{q} - m_t) = S^{-1}(q + k_g) - S^{-1}(q).$$

When all insertions of a longitudinal gluon into a closed fermion loop are summed with a common loop-momentum routing, the inverse propagator differences in Eq. (4.1) telescope:

$$k_g^\mu \sum_{\text{insertions}} \mathcal{H}_{\mu\nu}^{(\text{insertion})} = 0.$$

In the p_1 -soft-collinear endpoint,

$$k_{g1} \longrightarrow -p_1, \quad k_{g1}^2 \longrightarrow 0,$$

so the gluon momentum contracted into the CT numerator becomes an on-shell longitudinal momentum. If Eq. (4.1) were realized locally inside the grouped integrand, then

$$k_{g1}^{\mu_1} \sum_{\text{Ward orbit}} \mathcal{H}_{\mu_1\mu_2} = \mathcal{O}(\lambda),$$

and the first CT would scale like

$$|\mathcal{I}_{C_1}^{\text{Ward summed}}| \sim \lambda^{-4}$$

instead of λ^{-5} . The same argument applies to C_2 in the p_2 -soft-collinear endpoint. This is why restoring the local Ward orbit was a plausible rescue strategy.

4.2 Routing strategy that was implemented

The routing attempt used the following explicit strategy.

1. Set the defining light loop momentum K_0 by assigning `lmb_id=0` to the massless bridge edge between the two gluon insertions. This makes $k_1 = K_0$ directly, so the collinear approach can be made with K_0 parallel to the beam direction.
2. Use the exact- ξ helper-edge topology for the CTs. The original graphs carry the same helper external half-edges, but all helper edges are dummy there, so the original graph is independent of $Q(2), Q(3), Q(4), Q(5)$. The p_1 -collinear CT uses the active pair $Q(2), Q(4)$ and is independent of $Q(3), Q(5)$. The p_2 -collinear CT uses the active pair $Q(3), Q(5)$ and is independent of $Q(2), Q(4)$. This lets all active helpers be set directly to the same chosen ξ .

3. For the Ward-complete test, disable final-state symmetrization and retain the full set of Higgs orderings. Put all original graphs and all CTs into a single `group_id`, so the local sum has access to the full Ward orbit rather than only a final-state-symmetrized subset.
4. Pin the heavy-loop momentum K_1 by assigning `lmb_id=1` to a heavy top edge adjacent to a fixed labelled Higgs insertion. In the exact- ξ external ordering the physical Higgs labels are shifted by the helper externals, so the raw anchor H_2 is represented as the emitted external H_6 . The selected edge is one of the two heavy-loop edges adjacent to this labelled insertion, with the side chosen canonically from the neighboring labels and the orientation chosen consistently with the fermion flow.
5. Keep both fermion-flow orientations, i.e. for every Higgs ordering include the orientation in which the anchored momentum flows along the fermion flow and the orientation in which it flows against it. This is needed because the Ward sum telescopes only after the routed massive-loop momenta are aligned consistently across the orbit.

This routing improves the formal conditions for Eq. (4.1), but the direct four-dimensional scans still show the same endpoint hierarchy: the soft-collinear CT scales one power worse than the original. The conclusion from this check is that the current grouped integrand still does not realize the required local Ward cancellation strongly enough to remove the endpoint growth.

Chapter 5

Current Interpretation

The endpoint issue is not caused by a missing threshold subtraction in the four-dimensional object, nor by an explicit ratio inserted in the numerator. It is already visible in the purely local tensor numerator and the intended denominator structure. The problematic ingredients are:

- the CT denominator retains the two small light denominators k_1^2 and $k_{g_i}^2$ and replaces the other light denominator by $A_\xi \sim \lambda$;
- the original graph has a soft bridge numerator $k_1 \sim \lambda$;
- the CT numerator has a longitudinal contraction $k_{g_i}^\mu \mathcal{H}_{\mu\nu}$ that is finite unless the local Ward sum is realized;
- the final-state-symmetrized minimal groups do not contain the full massive-loop Ward orbit, and the Ward-complete routing test has not yet produced the required local telescoping cancellation.

Thus the soft-collinear endpoint has the structure

$$\frac{N_{\text{orig}} \sim \lambda}{D_{\text{orig}} \sim \lambda^5} \quad \text{versus} \quad \frac{N_{C_i} \sim 1}{D_{C_i} \sim \lambda^5}.$$

In 3D CFF variables this same mismatch appears as a one-power stronger endpoint growth of the CT weight.

Chapter 6

Idea: Symmetrized Collinear Counterterms

A possible way forward is to modify the local CT by a symmetrization that keeps its integrated value zero. Let $\Delta_i(\ell)$ denote one of the local collinear CT integrands, where ℓ denotes all loop variables. The current construction has

$$\int d\ell \Delta_i(\ell) = 0$$

after the corresponding integrated contribution is accounted for in the subtraction scheme. Consider a finite set S of measure-preserving maps σ acting on the local loop variables, external-Higgs labels, heavy-loop orientation, or beam/auxiliary- ξ assignments. Define

$$\Delta_i^{\text{sym}}(\ell) = \frac{1}{|S|} \sum_{\sigma \in S} J_\sigma(\ell) \Delta_i(\sigma(\ell)),$$

where J_σ is the Jacobian of the transformation. If the maps are bijective and cover the same integration domain, then

$$\int d\ell \Delta_i^{\text{sym}}(\ell) = \frac{1}{|S|} \sum_{\sigma \in S} \int d\ell_\sigma \Delta_i(\ell_\sigma) = 0.$$

The hope is to choose S so that the finite longitudinal pieces of $k_{g_i} \cdot \mathcal{H}$ cancel locally in the soft-collinear endpoint, while the strict collinear limit that the CT is designed to subtract is left unchanged.

Natural candidates for S include:

- permutations of the identical Higgs insertions on the massive loop;
- reversal of the massive-loop orientation, paired with the corresponding ordered Higgs labels;
- exchange of the two sides of the light bridge, paired with $p_1 \leftrightarrow p_2$ and $(Q(2), Q(4)) \leftrightarrow (Q(3), Q(5))$;
- a Ward-orbit average over all insertions of the longitudinal gluon on the massive loop, written as an explicitly symmetrized CT numerator rather than relying on the graph grouping alone.

This would not be a change in the integrated subtraction if the maps are implemented as exact changes of variables. It would instead redistribute the zero-integral CT locally to make the endpoint behavior compatible with the massive-loop Ward identity.

Appendix A

Full GammaLoop Drawing Grid

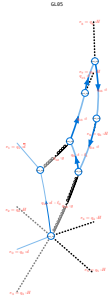
The following page is the direct output of running `just draw` in the GammaLoop `save dot` export of the final-state-symmetrized process.

Amplitudes

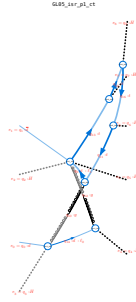
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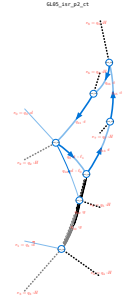
Graph #1



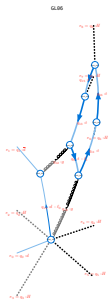
Graph #2



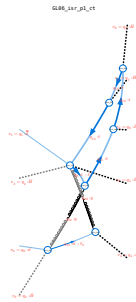
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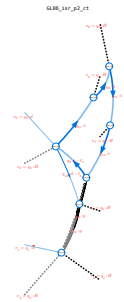
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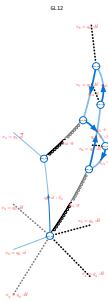
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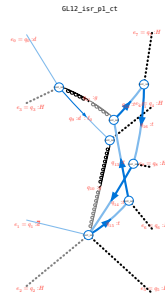
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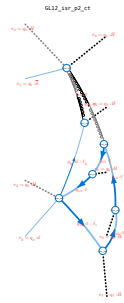
Graph #7



Graph #8



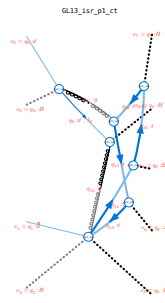
Graph #9



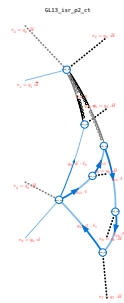
Graph #10



Graph #11



Graph #12



Bibliography

- [1] Kermanschah et al., *Local initial-state collinear subtraction for loop amplitudes*, [arXiv:2603.03169](#).